# 🔐 SOC Internship – Day 18 Notes

### Topic: Malware Analysis Basics (SOC Perspective)

### 🎯 Objective

Understand how SOC analysts detect, analyze, and respond to malware activity in a system.

## 🦠 1. What is Malware?

Malware (Malicious Software) is any software designed to:

- Damage systems
- Steal sensitive data
- Disrupt operations

## ⚠️ 2. Types of Malware

- **Virus** → Infects files and spreads when executed
- **Trojan** → Disguises as legitimate software
- **Ransomware** → Locks files and demands payment
- **Spyware** → Secretly monitors user activity

## 🚨 3. Indicators of Malware (IOCs)

SOC analysts look for:

- Unknown or suspicious processes running
- Connections to unusual or malicious IP addresses
- High CPU or memory usage
- Unauthorized file/system changes

## 🔍 4. Example Scenario

- Process: malware.exe
- Connected to IP: 45.23.11.9
- Created files in system directories

👉 These are strong signs of malicious behavior.

## 🧪 5. Malware Analysis Steps

1. Identify the suspicious process
2. Check network activity (IP connections)
3. Analyze system/file changes

## 🛡️ 6. Incident Response Actions

- Isolate the infected system
- Kill the malicious process
- Remove the malware completely

## 📝 Assignment Answers (Ready to Use)

### ✔ Explain Malware

Malware is harmful software designed to damage systems, steal data, or disrupt operations.

### ✔ Identify Indicators

- Unknown processes
- Suspicious IP connections
- High resource usage
- Unauthorized system changes

## ✔ Describe Response

- Isolate system
- Stop malicious process
- Remove malware and secure system